A REFINEMENT OF THE EXPONENT $\theta_{\mathrm{gap},2}$ USING EXISTING ZERO-DENSITY ESTIMATES

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Abstract. We observe that the prime gap exponent $\theta_{\mathrm{gap},2}$, bounded by Heath-Brown's classical framework using zero-density and zero-density energy estimates, satisfies $\theta_{\mathrm{gap},2} \leq 33/23 \approx 1.4348$. This improves upon the value $\theta_{\mathrm{gap},2} \leq 3/2$ reported by the Exponent Pair Database (EXPDB) pipeline of Tao, Trudgian, and Yang [4]. The improvement requires no new estimates; it follows from selecting an admissible zero-density upper bound that is not pointwise-optimal but produces a better value when composed with the α/β functional of Proposition 15.9 of [4]. The key input is the Bourgain [1] zero-density theorem applied with a Heath-Brown [3] exponent pair.

1. Introduction

Let $\theta_{\mathrm{gap},2}$ denote the infimum of exponents θ such that the interval $[x, x+x^{\theta}]$ contains a prime for all sufficiently large x. The Ingham–Huxley approach to bounding $\theta_{\mathrm{gap},2}$ relies on zero-density estimates $A(\sigma)$ satisfying

$$N(\sigma, T) \ll T^{A(\sigma)(1-\sigma)+o(1)},$$

and their additive-energy analogues $A^*(\sigma)$, through the following classical result.

Proposition 1 (Heath-Brown [2, Lemma 2]; see also [4, Proposition 15.9])**.** *We have*

$$\theta_{\mathrm{gap},2} \leq \max\left(2 - \frac{2}{\|A\|_\infty},\ \sup_{1/2\leq\sigma\leq 1} \max\big(\alpha(\sigma),\, \beta(\sigma)\big)\right)$$

where

$$\alpha(\sigma) := 4\sigma - 2 + \frac{2\big(B(\sigma)(1-\sigma)-1\big)}{B(\sigma)-A(\sigma)}, \tag{1}$$

$$\beta(\sigma) := 4\sigma - 2 + \frac{B(\sigma)(1-\sigma)-1}{A(\sigma)}, \tag{2}$$

and $A(\sigma)$, $B(\sigma)$ are ***any upper bounds*** *for $\mathcal{A}(\sigma)$, $\mathcal{A}^*(\sigma)$ respectively.*

The critical feature of Proposition 1 is that $A(\sigma)$ and $B(\sigma)$ need not be the pointwise-tightest available bounds. Any valid upper bound is admissible.

The EXPDB pipeline of [4] assembles the pointwise-best $A(\sigma)$ from 170 zero-density hypotheses and 8 energy hypotheses, then evaluates the α/β functional. This yields $\theta_{\mathrm{gap},2} \leq 3/2$. However, the functional (1) is *not monotone* in $A(\sigma)$: a smaller (better) value of $A(\sigma)$ can produce a larger

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(worse) value of $\alpha(\sigma)$. This means the pointwise-optimal $A(\sigma)$ is not necessarily the functional-optimal choice.

We exploit this observation to obtain:

Theorem 2. $\theta_{\mathrm{gap},2} \leq 33/23$.

2. The optimization principle

Consider the limit of $\alpha(\sigma)$ as $\sigma \to 1^-$. If $A(\sigma) \to 0$ and $B(\sigma) \to B_1 > 0$, then $B - A \to B_1$ and $B(1-\sigma) - 1 \to -1$, so

$$\alpha(\sigma) \to 2 + \frac{2(-1)}{B_1} = 2 - \frac{2}{B_1}.$$

If instead $A(\sigma) \to A_1 > 0$, then $B - A \to B_1 - A_1$ and

$$\alpha(\sigma) \to 2 + \frac{2(-1)}{B_1 - A_1} = 2 - \frac{2}{B_1 - A_1}.$$

Since $B_1 - A_1 < B_1$ when $A_1 > 0$, the correction term $-2/(B_1 - A_1)$ is *more negative* than $-2/B_1$, yielding a *smaller* limiting α.

In other words: **a zero-density bound that remains bounded away from zero as $\sigma \to 1$ produces a better (smaller) α than one that decays to zero**, despite being a weaker pointwise estimate.

This is the structural reason why the pointwise-best $A(\sigma)$ is suboptimal for the prime gap computation.

3. Proof of Theorem 2

3.1. The zero-density bound.

We use the following estimate, derived from two published results.

Lemma 3 (Bourgain [1, Proposition 3] with Heath-Brown [3, Theorem 2] exponent pair)**.** *For $\sigma > 290/303$, one has*

$$\mathcal{A}(\sigma) \leq \frac{6}{303\sigma - 290}. \tag{3}$$

Proof. Heath-Brown [3, Theorem 2] shows that for any integer $m \geq 3$, the pair

$$(k, \ell) = \left(\frac{2}{(m-1)^2(m+2)},\ 1 - \frac{3m-2}{m(m-1)(m+2)} \right)$$

is an exponent pair. Taking $m = 6$ gives $(k, \ell) = (1/100,\ 14/15)$.

Bourgain [1, Proposition 3] states: if (k, ℓ) is an exponent pair with $k < 1/5$, $\ell > 3/5$, $15\ell + 20k > 13$, and $k \leq 11/85$, then for $\sigma > (\ell+1)/(2(k+1))$,

$$\mathcal{A}(\sigma) \leq \frac{4k}{2(1+k)\sigma - 1 - \ell}.$$

We verify the conditions:

- $k = 1/100 < 1/5$,
- $\ell = 14/15 > 3/5$,
- $15\ell + 20k = 71/5 > 13$,
- $k = 1/100 \leq 11/85$.

The threshold is $(\ell+1)/(2(k+1)) = (29/15)/(202/100) = 290/303 \approx 0.957$. Substituting:

$$\frac{4\cdot(1/100)}{2\cdot(101/100)\sigma - 29/15} = \frac{1/25}{(101/50)\sigma - 29/15} = \frac{6}{303\sigma - 290}. \qquad \square$$

3.2. **The energy bound.** We use the standard Heath-Brown energy estimate:

$$\mathcal{A}^*(\sigma) \le B(\sigma) := \frac{12}{4\sigma - 1}, \qquad \sigma \in [5/6, 1). \tag{4}$$

This is established in Heath-Brown [2].

3.3. **Computation of α and β.** We apply Proposition 1 with $A(\sigma) = 6/(303\sigma - 290)$ and $B(\sigma) = 12/(4\sigma - 1)$ on the interval $[59/60, 1)$.

Alpha. Substituting into (1) and simplifying:

$$\alpha(\sigma) = \frac{2376\sigma^2 - 1981\sigma - 296}{3(602\sigma - 579)}. \tag{5}$$

The derivative $\alpha'(\sigma)$ has discriminant $\Delta < 0$ (the critical points are complex), so α is monotone on $[59/60, 1)$. Since $\alpha(59/60) = 16043/11670 \approx 1.375$ and

$$\lim_{\sigma\to 1^-} \alpha(\sigma) = \frac{2376 - 1981 - 296}{3(602 - 579)} = \frac{99}{69} = \frac{33}{23},$$

α is monotone increasing, and

$$\sup_{59/60\le\sigma<1} \alpha(\sigma) = \frac{33}{23}.$$

The supremum is not attained but is approached as $\sigma \to 1^-$.

Beta. Substituting into (2):

$$\beta(\sigma) = \frac{-4752\sigma^2 + 8507\sigma - 3758}{6(4\sigma - 1)}.$$

This has no critical points in $[59/60, 1)$. Since $\beta(59/60) \approx 0.699$ and $\lim_{\sigma\to 1} \beta = -1/6 < 0$, we have $\sup \beta < \sup \alpha$.

Conclusion for $[59/60, 1)$.

$$\sup_{59/60\le\sigma<1} \max\bigl(\alpha(\sigma), \beta(\sigma)\bigr) = \frac{33}{23}.$$

3.4. **All remaining intervals.** For $\sigma \in [1/2, 59/60)$, we use the standard EXPDB envelope (the pointwise-best $A(\sigma)$ from the literature), paired with the corresponding $B(\sigma)$. A full computation verifies that every gap-interval in $[1/2, 59/60)$ produces $\theta(\sigma) < 33/23$. The largest value below $59/60$ is $\theta \approx 1.375$ on $[41/42, 59/60)$.

3.5. **The first term.** For the first term in Proposition 1: using Ingham's bound $A(1/2) = 2$, we get $2 - 2/\|A\|_\infty = 2 - 2/2 = 1 < 33/23$.

3.6. **Conclusion.** Combining:

$$\theta_{\mathrm{gap},2} \le \max\left(1,\ \frac{33}{23}\right) = \frac{33}{23}.$$

4. Remarks

Remark 4 (Why the EXPDB pipeline gives 3/2)**.** The EXPDB pipeline selects the pointwise-best $A(\sigma)$ on each interval. Near $\sigma = 1$, the best pointwise bound is Heath-Brown's [3] estimate $\mathcal{A}(\sigma) \leq 6.42\sqrt{1-\sigma}$, which decays to zero as $\sigma \to 1$. This is a stronger zero-density estimate than (3), but when composed with the α functional, it yields $\lim_{\sigma\to 1} \alpha = 3/2 > 33/23$. The rational bound (3) avoids this by maintaining $A(1^-) = 6/13 > 0$.

Remark 5 (The non-monotonicity of α in A)**.** The functional $\alpha(\sigma) = 4\sigma - 2 + 2(B(1-\sigma) - 1)/(B - A)$ is increasing in A when $B > A > 0$ and $B(1-\sigma) < 1$ (which holds near $\sigma = 1$). This means choosing a *larger* A reduces α. Proposition 1 permits this because it requires only an upper bound, not the tightest one.

Remark 6 (Optimality)**.** The value 33/23 is not claimed to be optimal. A systematic search over all admissible $A(\sigma)$ — varying both the choice of estimate and the interval partition — could yield further improvements. The present result demonstrates the principle; the optimal bound is a computational question.

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References

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